

Mahmoud Aziz Ben Khalifa

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PROFILE

Final-year engineering student at EIDD, Université Paris Cité, seeking a final-year internship (PFE) in AI and embedded systems. Experience in speech processing, computer vision and autonomous robotics.

PROFESSIONAL EXPERIENCE

Université du Québec à Trois-Rivières (UQTR), Canada May - July 2026

Mitacs Research Intern | AI & Speech Processing

- Developed a Mandarin-to-English speech-to-speech translation pipeline without intermediate text during training.
- Integrated MAST wavelet features and self-supervised speech representations into neural translation models; implemented evaluation and improved pipeline reliability.

CNN · Transformer · MAST · w2v-BERT 2.0 · HuBERT · Signal processing · Deep learning

LA.R.A Automation Laboratory, ENIT, Tunisia

June - July 2025

Robotics Intern | Autonomous Mobile Robot Control

- Implemented fuzzy and neuro-fuzzy controllers on STM32 with FreeRTOS; fused ultrasonic, gyroscope and wheel encoder data for robot trajectory estimation.

STM32 · FreeRTOS · Sensor fusion · Fuzzy control

ACT Nexans Autoelectric, Tunisia

July 2024

Technical Intern | Electrical Test Systems

- Maintained automotive wiring test benches using IVIS5, Ivision and CsWin.

EDUCATION

EIDD | Université Paris Cité, France

2025 - 2027

Engineering Degree in Embedded Computer Systems | Final year, in progress.

National Engineering School of Tunis (ENIT), Tunisia

2023 - 2025

Engineering Degree in Electrical Engineering.

IPEIT | Preparatory Institute for Engineering Studies of Tunis

2021 - 2023

Preparatory Cycle for Engineering Studies.

Bourguiba Pioneer High School, Tunis

2021

Baccalaureate in Mathematics, with honors.

SELECTED PROJECTS

AI-Powered Highway Scene Analyzer

May 2026

Combined YOLOv8/YOLOv11 detection with a LLaMA 3.3 ReAct agent for road-risk assessment on BDD100K; built a Streamlit dashboard.

Vehicle Telematics Platform | Engineering Project

April 2026

Built GPS tracking and SOS alerts with Teltonika, Python, PostgreSQL, Node.js/Express and Leaflet maps.

Quiz Management Application | Java & JavaFX

March 2026

Built quiz creation and scoring interfaces with layered architecture, JUnit tests and GitHub Actions CI.

Cardiac Pathology Recognition | Final Year Project

May 2025

Developed image-based pathology recognition using deep learning, Python and NVIDIA Jetson Nano.

DC Motor Speed & Direction Controller

May 2024

Designed an electronic control board using Eagle to regulate motor speed and direction.

Autonomous Line-Following Robot

January 2024

Built an Arduino/C++ robot with IR/ultrasonic sensors, obstacle avoidance and PID control.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL, VHDL, MATLAB, HTML.

AI & signal processing: CNNs, Transformers, YOLO, LLaMA, ReAct agents, wavelets.

Software & data: JavaFX, Node.js/Express, REST APIs, PostgreSQL, Leaflet, Streamlit.

Testing & CI: Git/GitHub Actions, Maven, JUnit, Mockito, JaCoCo, SonarQube.

Embedded & electronics: STM32, Arduino, ESP32, Raspberry Pi, Jetson, FreeRTOS, PID, sensor fusion, STM32CubeIDE, Keil, Eagle, PSpice; Linux.

LANGUAGES & ACTIVITIES

French: C1 | **English:** B2 | **Arabic:** Native | **German:** Basic

Senior member, ENIT Electrical Engineering Club.